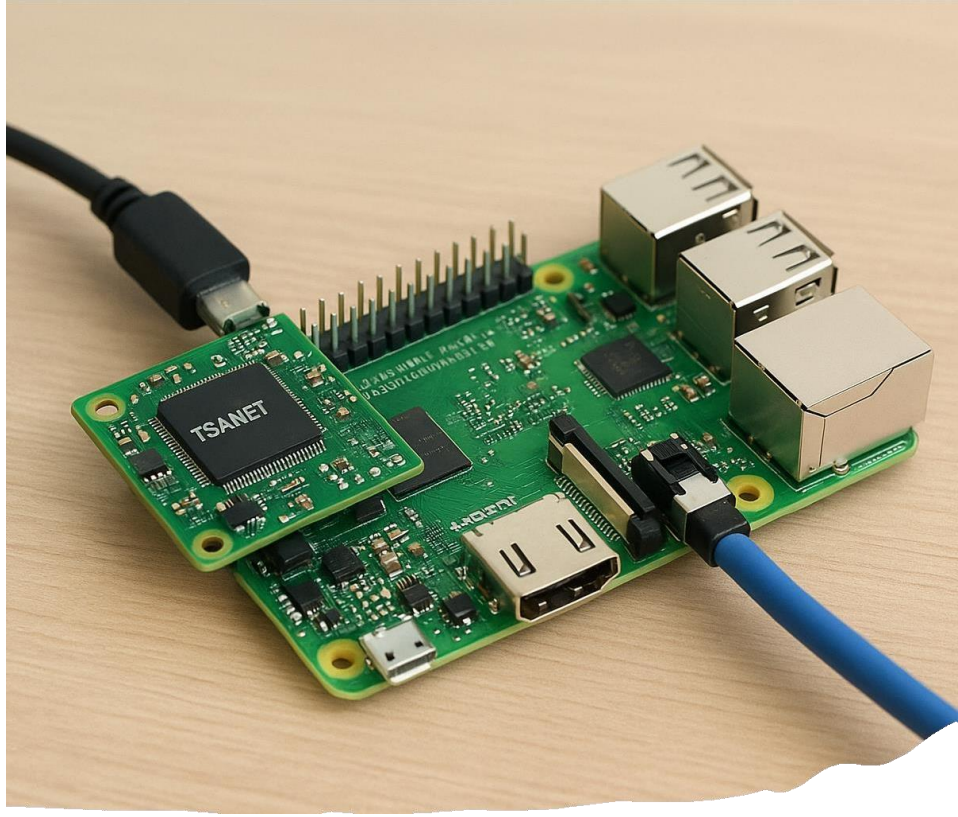

Lightweight Real-Time Intrusion Detection for IoT Networks Using TSANet Deployed on Raspberry Pi with Live Traffic Monitoring

PROFILE

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DEPT OF CS
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Introduction

- Increasing frequency and sophistication of cyberattacks (DoS, DDoS, brute force, Heartbleed etc.).
- IoT networks generate continuous traffic and are highly vulnerable to cyberattacks.
- Traditional IDS solutions depend on cloud or high-power systems and are not suitable for edge devices.
- This project introduces a **lightweight TSANet-based IDS** deployed entirely on a **Raspberry Pi 5**.

LITERATURE REVIEW

1. Paper Title : A Lightweight Optimized Deep Learning-based Host-Intrusion Detection System Deployed on the Edge for IoT

- CNN-based approach with 5-layer architecture (Conv1D, MaxPooling1D, Flatten, Output). **Edge deployment** on Raspberry Pi and microcontrollers. **Model optimization** through weights pruning, post-training quantization, and weight clustering. **TensorFlow Lite** conversion for lightweight deployment. Training: 10 epochs, batch size 32, parameters reduced from 26,365 to 645.
- **MQTT-IOT-IDS2020 dataset** (uniflow features version). Generated from simulated MQTT network with 12 sensors, broker, camera, and attacker. **5 labels:** Normal traffic, Aggressive scan, UDP scan, Sparta SSH brute-force, MQTT brute-force attack. Data split: 60% training, 20% validation, 20% testing.

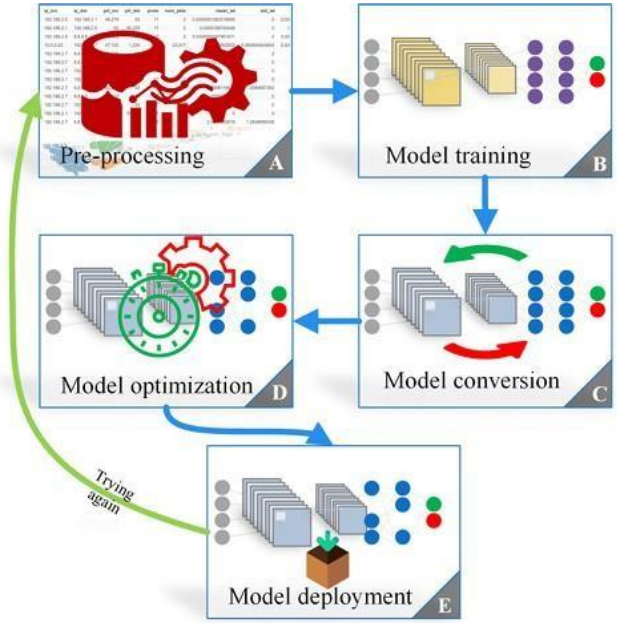


Figure 4. DL-IDS preparation and deployment steps

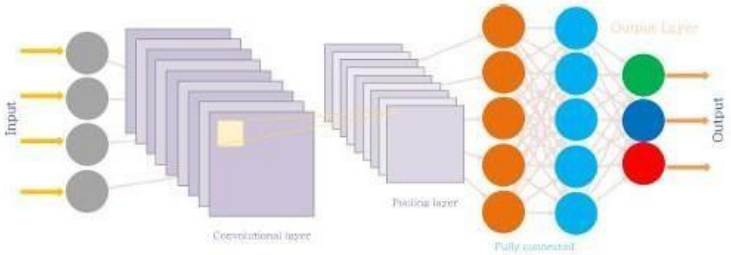


Figure 1. CNN architecture

- **Accuracy:** 96.69% (optimized model), 99.74% (original model). **Model size:** Reduced from 343 KB to 2704bytes. **Memory usage:** Significantly reduced for edge deployment. **Inference time:** Faster due to edgeprocessing.
- **Generalization challenge:** Need for device-specific customization rather thanuniversal solution. **Resource constraints:** Limited by IoT device computational capabilities.
- **Scalability:** Difficulty in standardizing across diverse IoThardware.

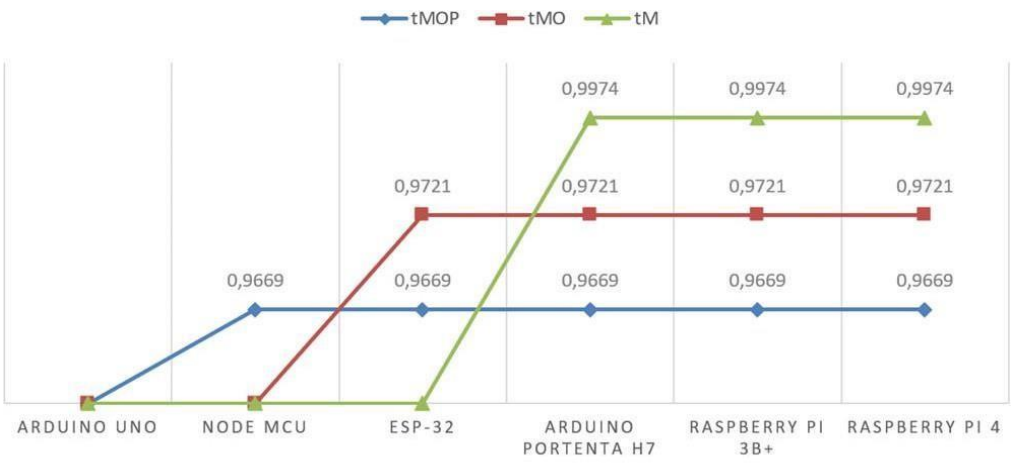


Figure 7. The suited model for each device

LITERATURE REVIEW

2. Paper Title : Anomaly-based Intrusion Detection System for IoT Networks through Deep Learning Model

- **CNN-based anomaly detection model.** Sequential CNN with categorical/binary cross-entropy loss, Adam optimizer, 0.5 dropout, 0.2 test size, batch size 10, trained for 30 epochs. **Deep learning** for traffic analysis to classify normal vs abnormal behavior. **Parameter sharing** and sparse connections used for scalability.
- **NID dataset:** 41 TCP/IP features, 25,192 training records, 5,039 validation records (20%). **BoT-IoT dataset:** Multi-class with DoS, DDoS, reconnaissance, and normal traffic (UNSW Cyber Range Lab). Train-test split: 80–20%. Includes various attack types and subcategories.

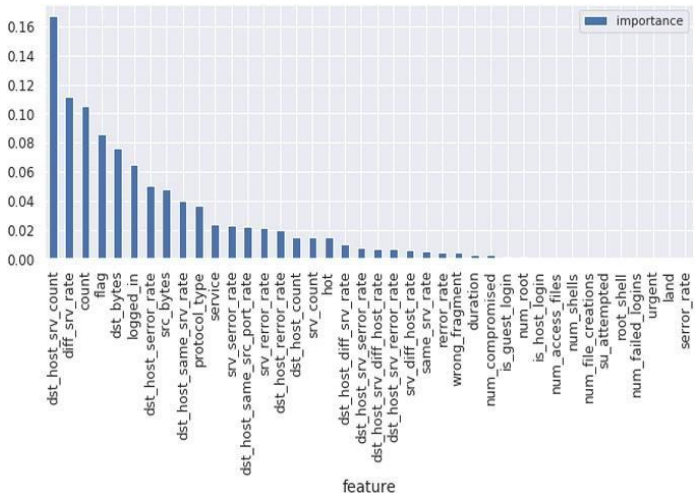


Figure 4: Features of the NID dataset

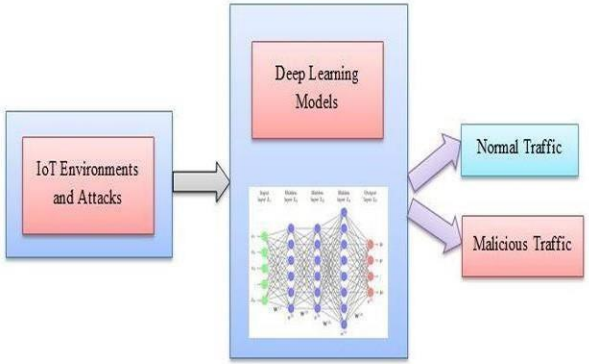


Figure 3. IDS using deep learning models

- **Traditional IDS limitations:** Not secure for IoT due to resource constraints. **High computational requirements:** Needs power and storage. **False positives:** Anomaly-based IDS prone to misclassification. **Dataset issues:** KDDCup99 lacks homogeneous IoT-specific data.
- **Accuracy:** 99.51% (NID), 95.55% (BoT-IoT). **Class-specific accuracy:** DDoS(86.18%), DoS (99.97%), Normal (100%), Reconnaissance (100%).**Confusion matrix analysis** and training/validation accuracy curvespresented.
- **IoT-specific security:** Need lightweight IDS for constrained devices. **Energy limitations:** IoT devices cannot run advanced IDS. **Unattended operation:** Vulnerability to malicious access and eavesdropping. **Dataset validity:** Need benchmark IoT-specific datasets for fair evaluation.

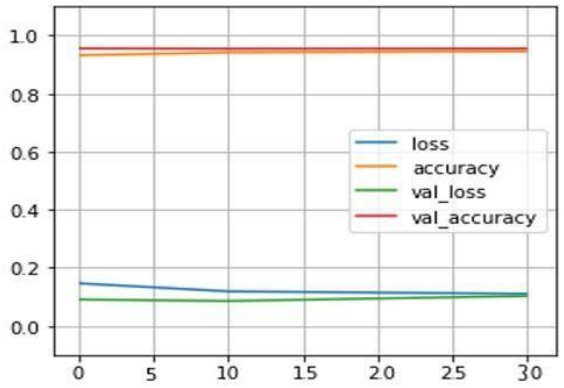
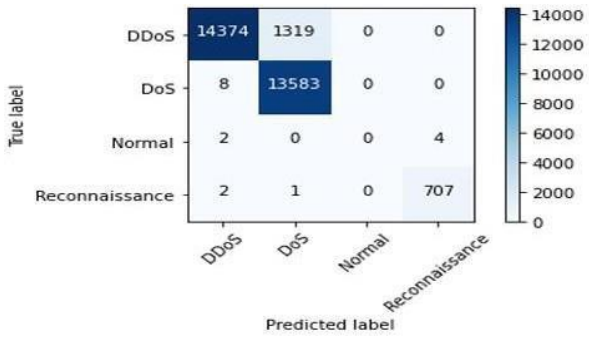


Figure 8: Training and validation accuracies on BoT-IoT dataset



LITERATURE REVIEW

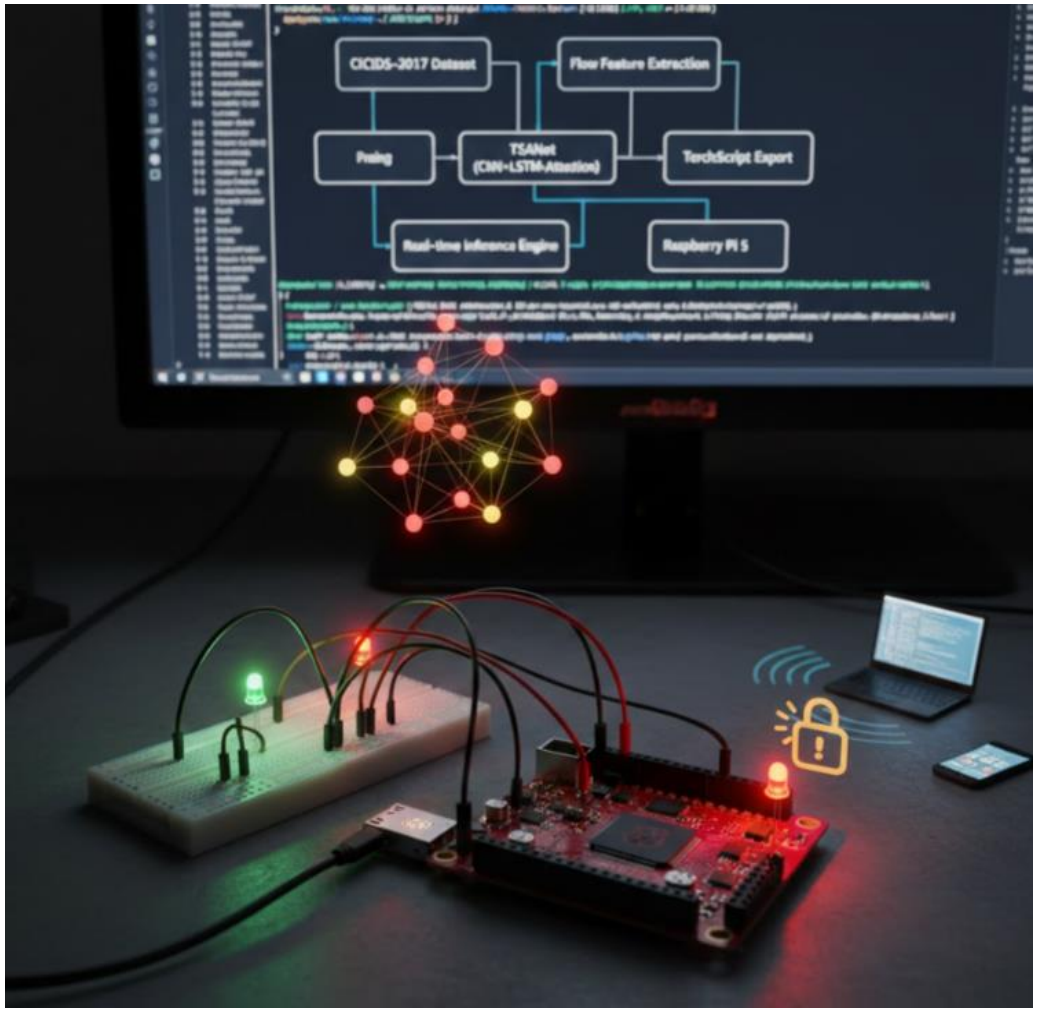
Paper Title	Methodology/ Implementation	Data set Description	Performance Parameter	Limitations/ Disadvtanges	Gaps Identified
3.Deep-IDS: A Real-Time Intrusion Detector for IoT Nodes Using Deep Learning	LSTM-based model, real-time blocking at 70% threshold, Raspberry Pi 4 (TensorFlow).	CIC-IDS2017, 56k cleaned records, 78 features, 15 classes.	Accuracy 97.67%, class-specific >93%.	Needs Pi 4 with 4GB RAM, threshold fixed, possible delays.	Limited dataset, real-time trade-off, deployment on diverse IoT devices.
4.Design and Implementation of a Machine Learning-Based Adaptive IDS on Raspberry Pi for Smart Home Network Security	Adaptive ML on Raspberry Pi, real-time logging, modular framework.	Simulated smart home data, DDoS, brute force, malware injection.	Accuracy 85%, response time 2s, low power use.	Limited Pi resources, performance drop with heavy loads.	Need real-world validation, scalability for larger networks.

PROBLEM STATEMENT

IoT networks generate continuous, diverse, and often unpredictable traffic, yet most existing Intrusion Detection Systems (IDS) rely on heavy deep-learning models that cannot run on resource-constrained edge devices like the Raspberry Pi. Centralized cloud-based IDS solutions introduce latency, privacy risks, and bandwidth overhead, while many lightweight IDS approaches fail to capture the temporal and spatial behaviour of real attacks. Moreover, most research is evaluated only on offline datasets and not on live Wi-Fi traffic, making them impractical for real deployments.

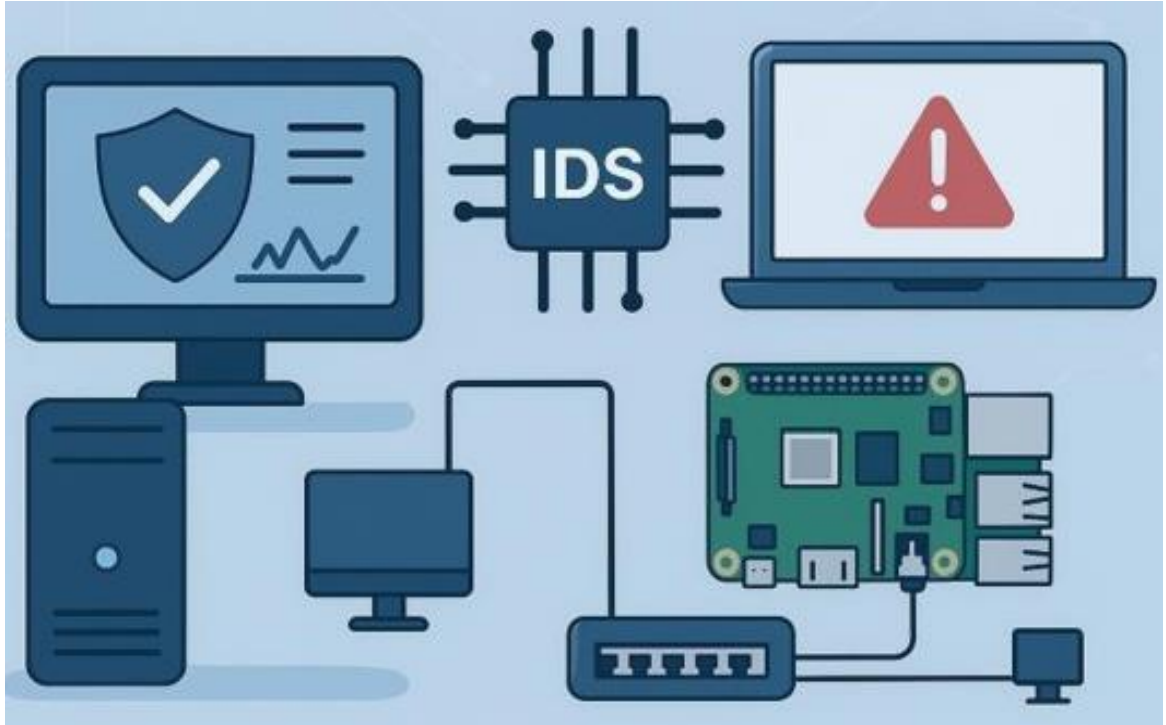
PROPOSED SOLUTION

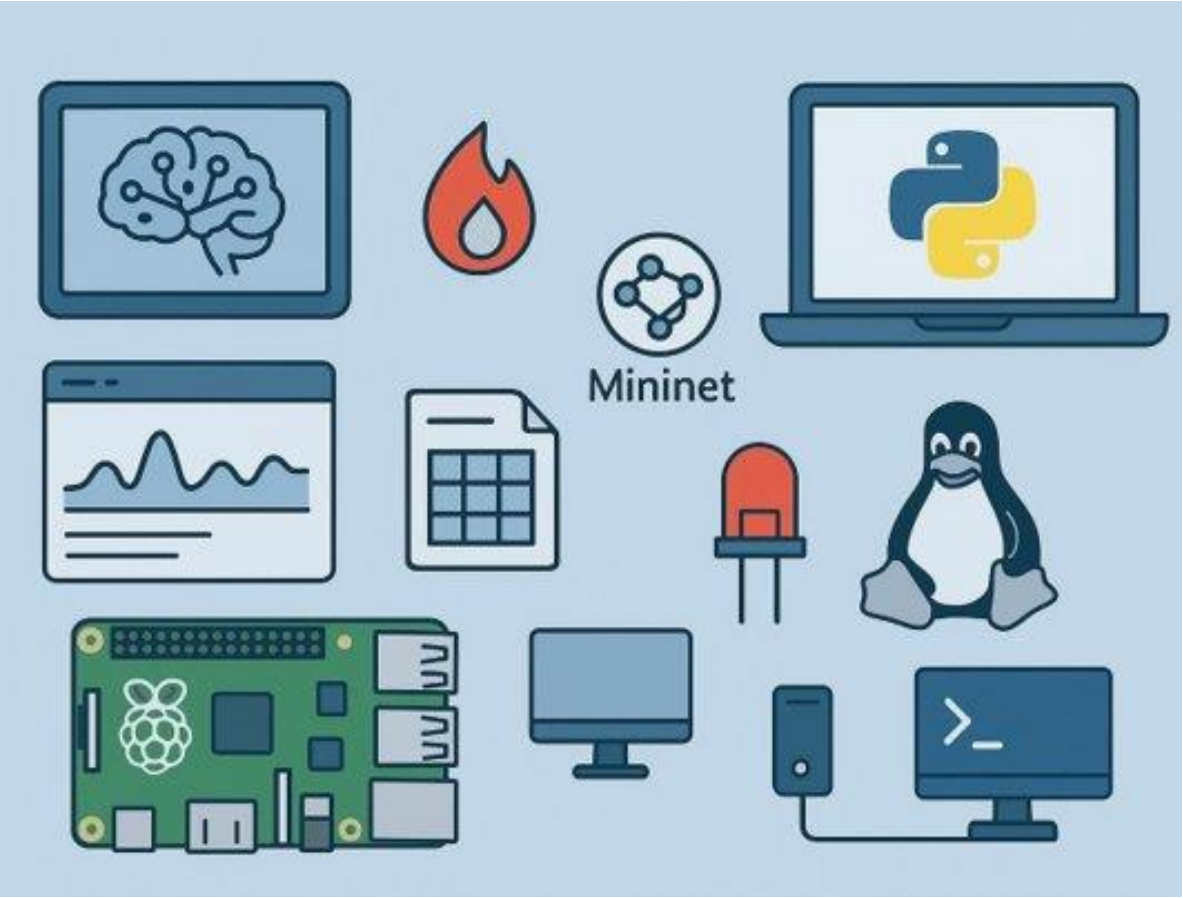
- Train a lightweight deep-learning model (TSANet) using 32 engineered flow features from the CICIDS-2017 dataset.
- Use a hybrid architecture combining CNN, LSTM, and attention mechanisms to capture both spatial and temporal behavior in network flows.
- Optimize the trained model through pruning and quantization to reduce size and latency while maintaining accuracy.
- Export the optimized model to TorchScript to ensure compatibility with ARM-based edge devices.
- Deploy the model on a Raspberry Pi 5 configured as a real-time inference engine for intrusion detection.
- Configure the Raspberry Pi as a Wi-Fi access point to capture live traffic, reconstruct flows, extract features, and run the TSANet model directly on-device.
- Enable real-time alerts using GPIO-based LED outputs for detected attacks.



OBJECTIVES

- To design and train a lightweight deep learning–based intrusion detection model (TSANet) using the CICIDS-2017 dataset.
- To optimize the trained model through pruning and quantization for efficient edge deployment.
- To convert the optimized model into TorchScript for execution on ARM-based devices.
- To deploy the IDS on a Raspberry Pi 5 and enable real-time inference directly on the device.
- To configure the Raspberry Pi as a Wi-Fi access point for capturing genuine live network traffic.
- To perform real-time intrusion detection, including flow reconstruction, feature extraction, and alert generation on the Raspberry Pi.



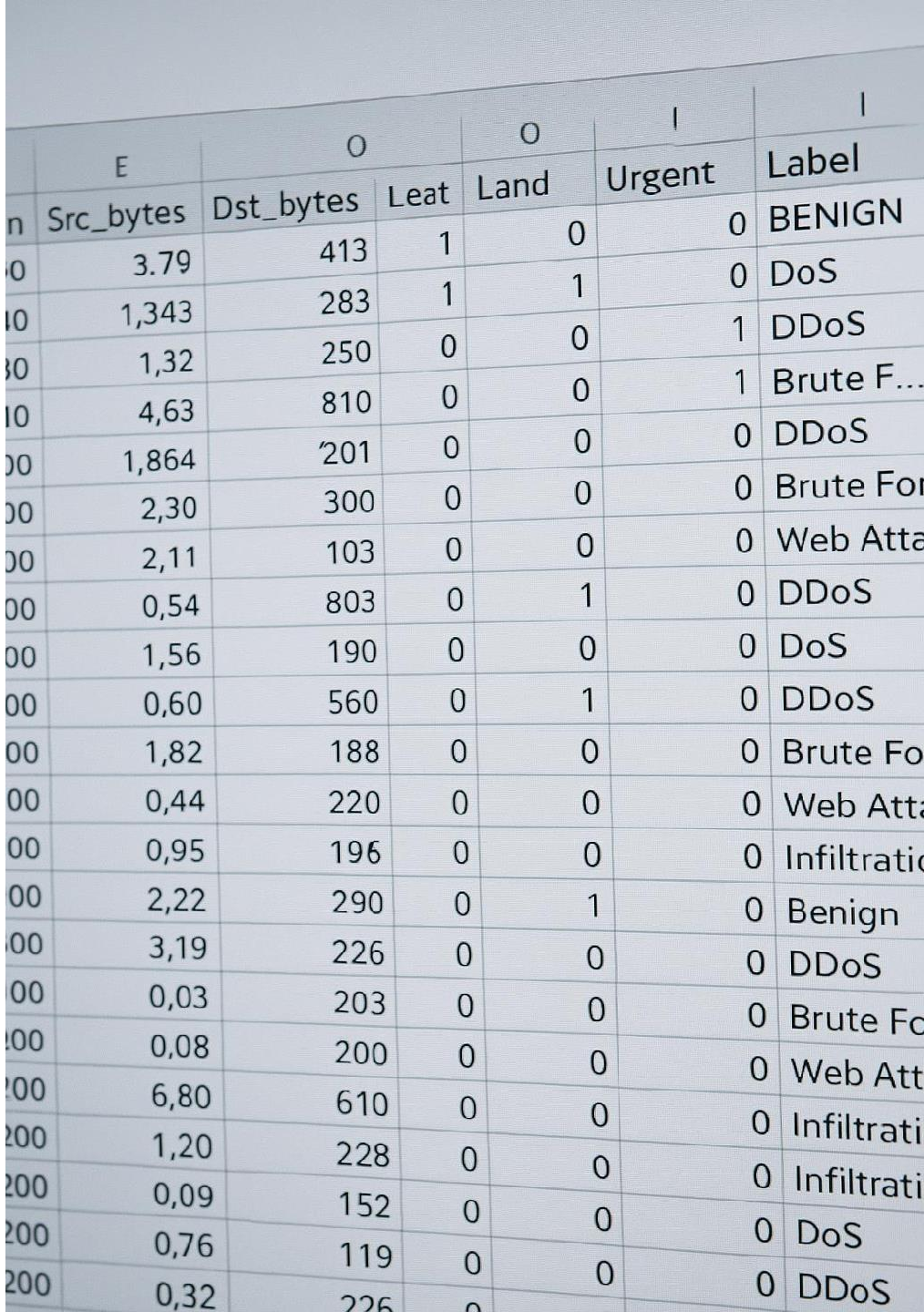


TOOL LEARNING

- Python 3.11
  Learned to build end-to-end data pipelines, preprocessing scripts, and real-time packet processing modules.
- PyTorch & TorchScript
  Trained deep learning models, applied pruning & quantization, and exported models for ARM-based deployment.
- Scapy
  Understood low-level packet sniffing, flow reconstruction, protocol fields, and live traffic parsing.
- Raspberry Pi OS & Linux Networking
  Configured interfaces, routing rules, DHCP, and system-level network services.
- hostapd & dnsmasq
  Learned to set up a Raspberry Pi as a fully functional Wi-Fi access point for real traffic capture.
- NumPy, Pandas, Joblib
  Used for dataset preprocessing, feature extraction, normalization, and offline testing.
- GPIO (RPi.GPIO)
  Implemented hardware alerts for attack detection using LEDs.
- VS Code / Terminal / SSH
  Real-time debugging, remote execution, and monitoring of the deployed IDS system.

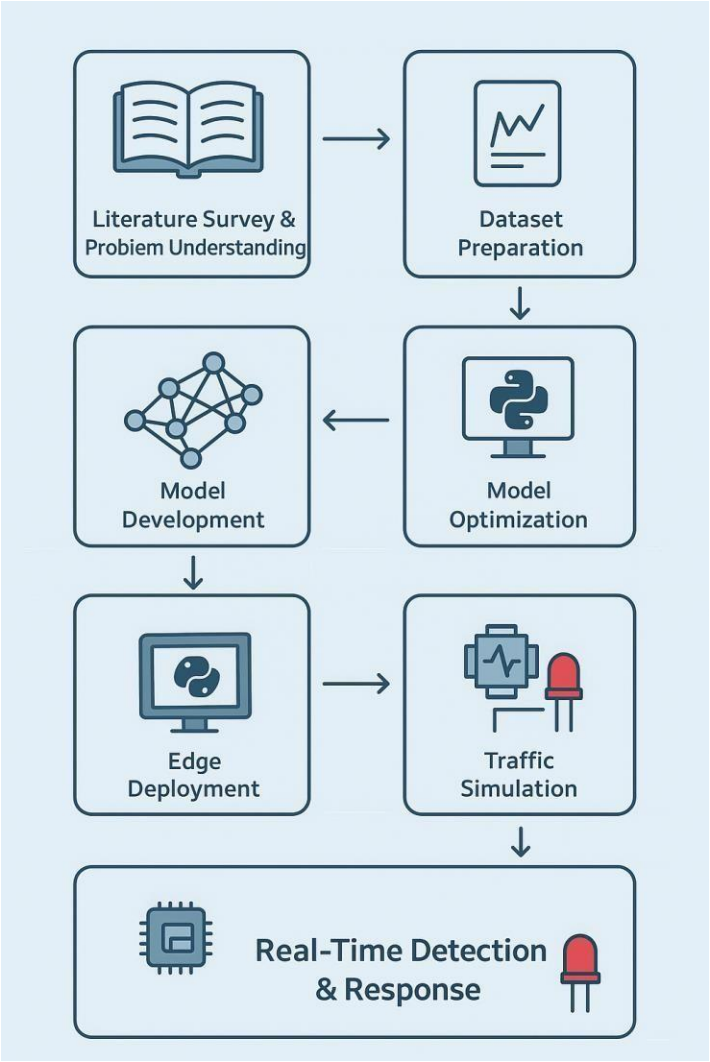
DATASET DESCRIPTION

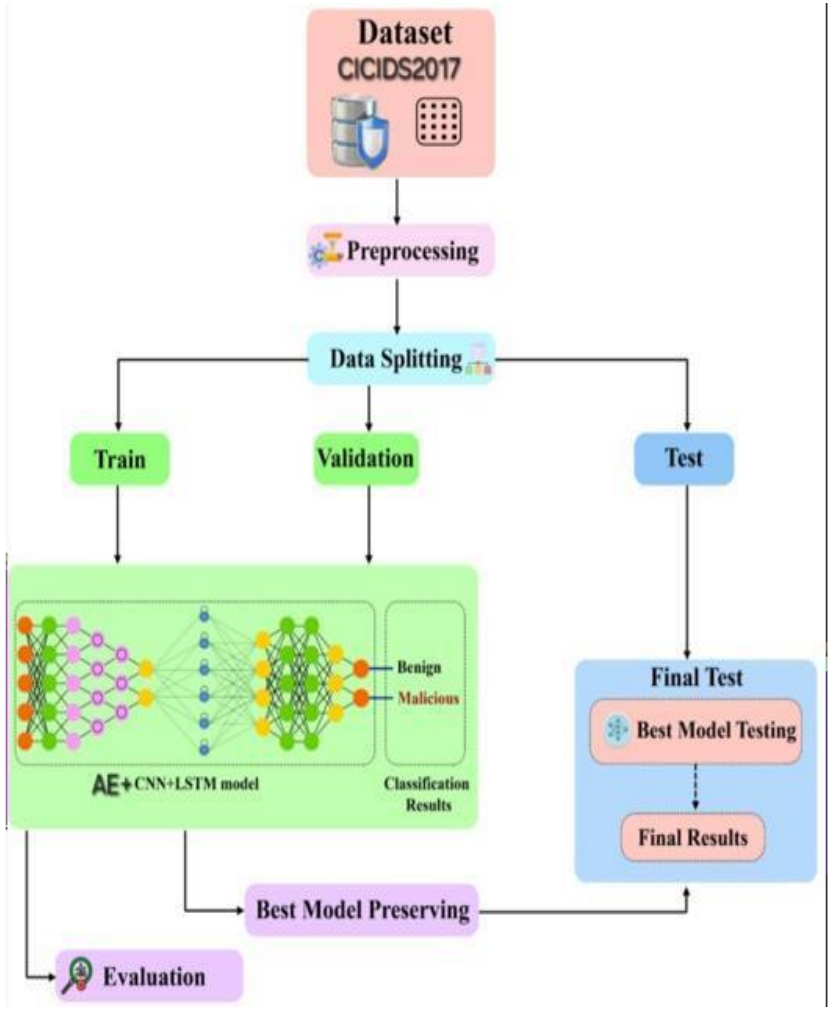
- The CICIDS2017 dataset is a benchmark dataset for intrusion detection, created by the Canadian Institute for Cybersecurity. It contains 8 CSV files, each representing traffic captured on different days and attack scenarios. Across all files, the dataset has 80 features (network flow characteristics such as packet counts, durations, byte rates, and flag counts) plus a Label column indicating *BENIGN* or specific attack types.
- Example file statistics:
- Thursday-WorkingHours-Morning-WebAttacks.pcap_ISCX.csv – ~230,000 rows, 79 feature columns +Label
- Thursday-WorkingHours-Afternoon-Infiltration.pcap_ISCX.csv – ~170,000 rows, 78 feature columns +Label
- The dataset includes both benign traffic and multiple attack categories such as DoS, DDoS, Brute Force, Web Attacks, and Infiltration.



	E	O		O	I	I
n	Src_bytes	Dst_bytes	Leat	Land	Urgent	Label
0	3.79	413	1	0	0	BENIGN
0	1,343	283	1	1	0	DoS
0	1,32	250	0	0	1	DDoS
0	4,63	810	0	0	1	Brute F...
00	1,864	201	0	0	0	DDoS
00	2,30	300	0	0	0	Brute For
00	2,11	103	0	0	0	Web Atta
00	0,54	803	0	1	0	DDoS
00	1,56	190	0	0	0	DoS
00	0,60	560	0	1	0	DDoS
00	1,82	188	0	0	0	Brute Fo
00	0,44	220	0	0	0	Web Atta
00	0,95	196	0	0	0	Infiltratio
00	2,22	290	0	1	0	Benign
00	3,19	226	0	0	0	DDoS
00	0,03	203	0	0	0	Brute Fo
00	0,08	200	0	0	0	Web Att
00	6,80	610	0	0	0	Infiltrati
00	1,20	228	0	0	0	Infiltrati
00	0,09	152	0	0	0	DoS
00	0,76	119	0	0	0	DDoS
00	0,32	226	0			

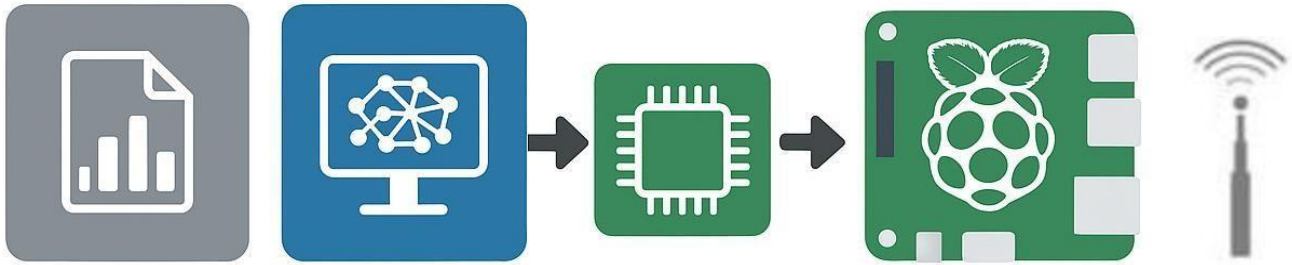
ROADMAP


Literature Survey & Probiem Understanding
Dataset Preparation
Model Development
Model Optimization
Edge Deployment
Traffic Simulation
Real-Time Detection & Response


Dataset
CICIDS2017
Preprocessing
Data Splitting
Train
Validation
Test
Benign
Malicious
AE+CNN+LSTM model
Classification Results
Final Test
Best Model Testing
Final Results
Best Model Preserving
Evaluation

IMPLEMENTATION

- **Dataset Preparation**
- **Model Design and Training**
- **Model Conversion for Deployment**
- **Deployment on Raspberry Pi**
- **Traffic Generation**



RESULTS

Full Feature Training Result:

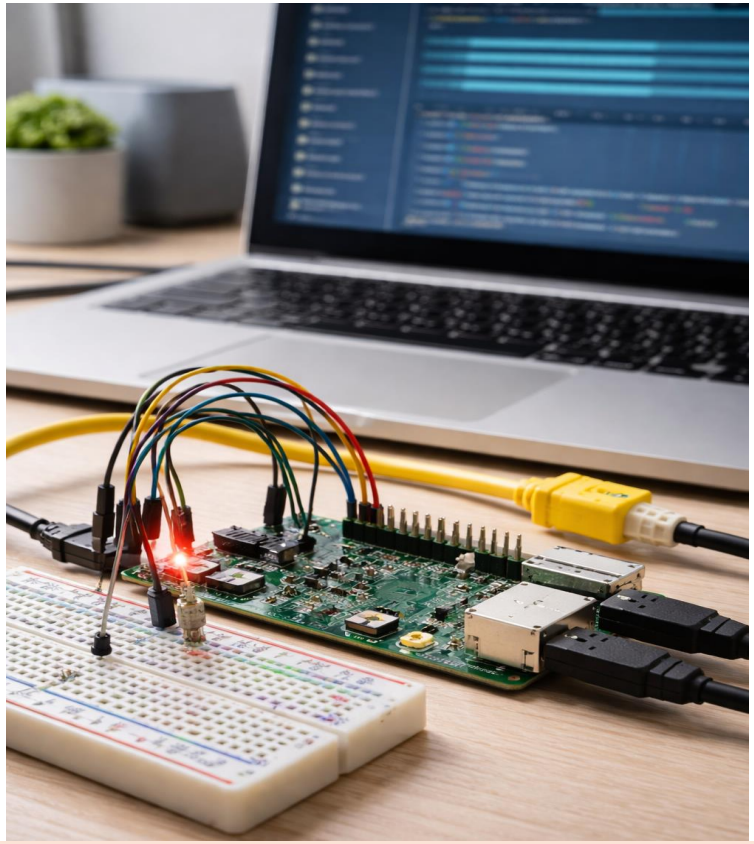
- Training Accuracy ≈ 98%
- Test Accuracy ≈ 97.35%

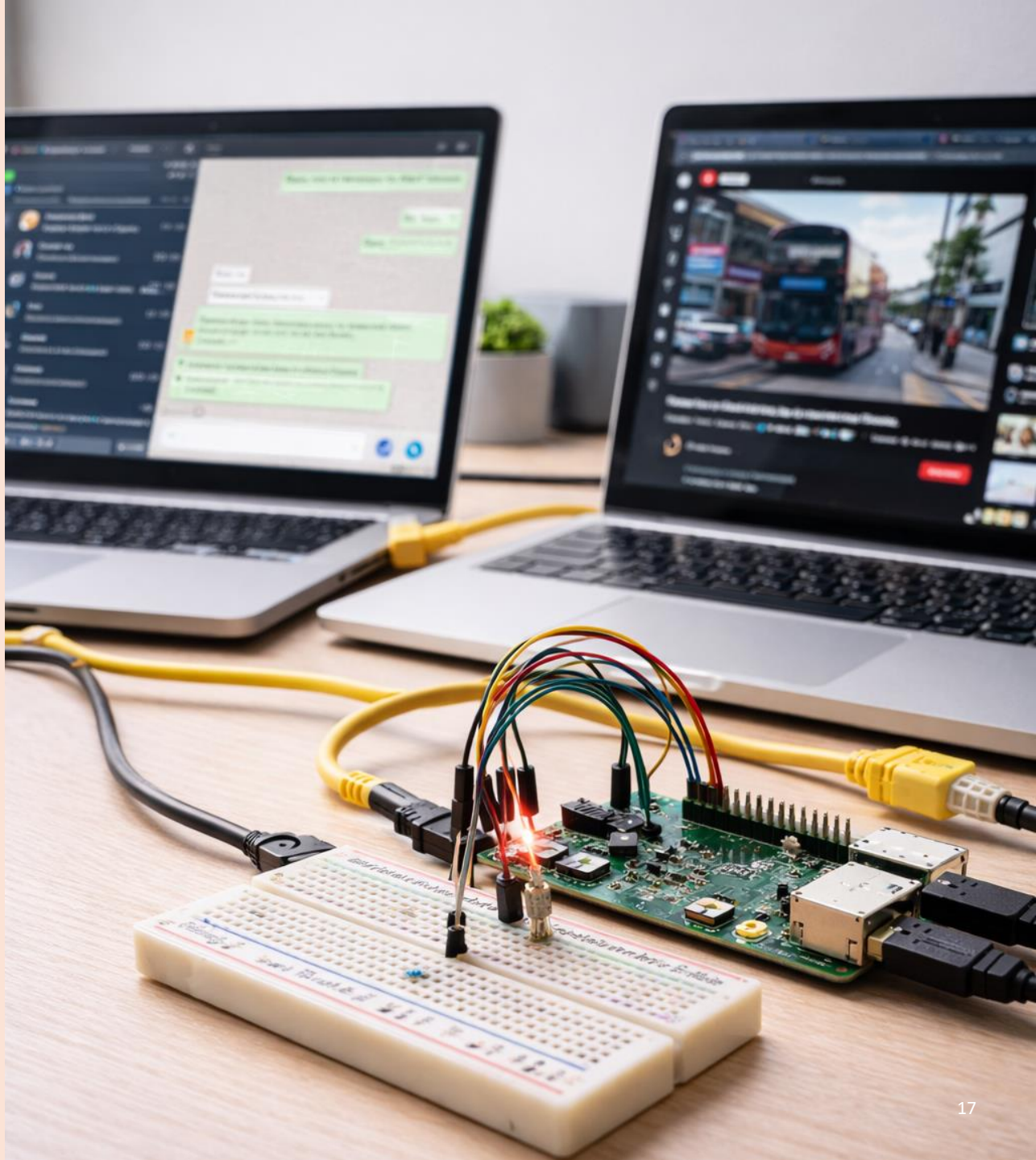
Model Optimization Result:

- Accuracy improved slightly to 98.3%
- Inference time reduced by ~44%.

```
Starting real-time IDS... press Ctrl+C to stop.
🚨 Prediction: BENIGN (conf=0.8604)
🚨 Prediction: BENIGN (conf=0.6664)
🚨 Prediction: Heartbleed (conf=0.4989)
🚨 Prediction: BENIGN (conf=0.9017)
🚨 Prediction: BENIGN (conf=0.8954)
🚨 Prediction: DoS (conf=0.4847)
🚨 Prediction: DoS (conf=0.7415)
🚨 Prediction: BENIGN (conf=0.8859)
🚨 Prediction: DoS (conf=0.5000)
🚨 Prediction: BENIGN (conf=0.9593)
🚨 Prediction: BENIGN (conf=0.9117)
🚨 Prediction: BENIGN (conf=0.9123)
🚨 Prediction: BENIGN (conf=0.5800)
🚨 Prediction: Heartbleed (conf=0.6001)
🚨 Prediction: Heartbleed (conf=0.4682)
🚨 Prediction: BENIGN (conf=0.5594)
🚨 Prediction: BENIGN (conf=0.8834)
🚨 Prediction: BENIGN (conf=0.8825)
🚨 Prediction: Heartbleed (conf=0.5531)
🚨 Prediction: BENIGN (conf=0.8912)
🚨 Prediction: BENIGN (conf=0.5482)
🚨 Prediction: BENIGN (conf=0.8099)
🚨 Prediction: BENIGN (conf=0.9201)
🚨 Prediction: Heartbleed (conf=0.7096)
...
```

HARDWARE SETUP





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🚦 Starting real-time IDS... press Ctrl+C
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🚨 Prediction: BENIGN (conf=0.9017)\n"
🚨 Prediction: BENIGN (conf=0.8954)\n"
🚨 Prediction: DoS (conf=0.4847)\n"
🚨 Prediction: DoS (conf=0.7415)\n"
🚨 Prediction: BENIGN (conf=0.8859)\n"
🚨 Prediction: DoS (conf=0.5000)\n"
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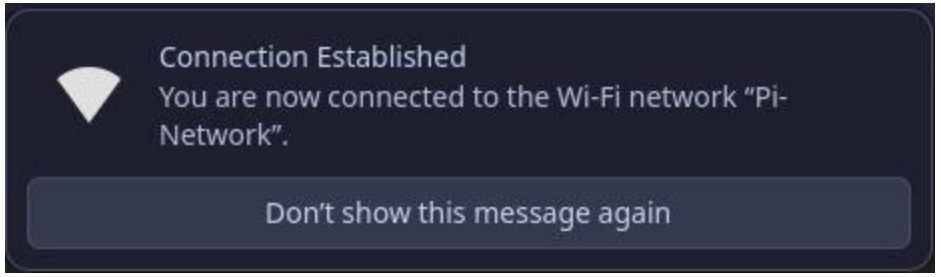
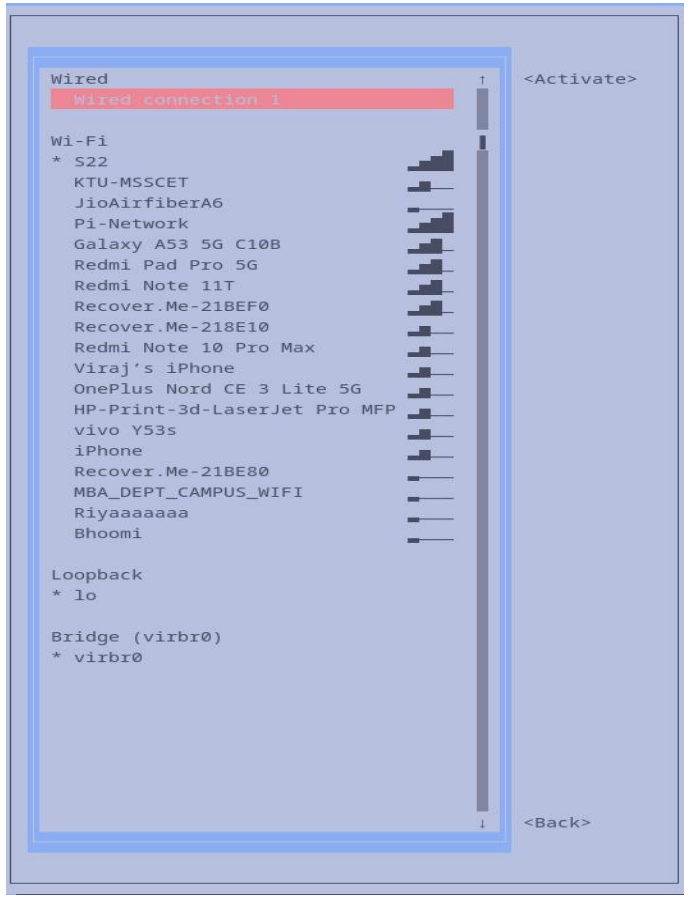
🚨 Prediction: BENIGN (conf=0.8825)\n"
🚨 Prediction: Heartbleed (conf=0.5531)\n"
🚨 Prediction: BENIGN (conf=0.8912)\n"
🚨 Prediction: BENIGN (conf=0.5482)\n"
🚨 Prediction: BENIGN (conf=0.8099)\n"
🚨 Prediction: BENIGN (conf=0.9201)\n"
🚨 Prediction: Heartbleed (conf=0.7096)\n"
🚨 Prediction: BENIGN (conf=0.8419)\n"
🚨 Prediction: DoS (conf=0.7487)\n"
🚨 Prediction: BENIGN (conf=0.8904)\n"
🚨 Prediction: DoS (conf=0.4722)\n"
🚨 Prediction: BENIGN (conf=0.8908)\n"
🚨 Prediction: BENIGN (conf=0.9222)\n"
🚨 Prediction: DoS (conf=0.7348)\n"
🚨 Prediction: BENIGN (conf=0.5535)\n"
🚨 Prediction: BENIGN (conf=0.9241)\n"
```

Results






```
pi@raspberrypi:~ $

========== ====== ALERT ======
Flow: ('163.70.144.8', 443) -> ('192.168.50.100', 45190)
Prediction: BENIGN (conf=0.5827)
==================

========== ====== ALERT ======
Flow: ('192.168.50.140', 8080) -> ('192.168.50.33', 41076)
Prediction: Heartbleed (conf=0.7354)
==================

========== ====== ALERT ======
Flow: ('192.168.50.140', 52550) -> ('34.107.221.82', 80)
Prediction: BENIGN (conf=0.5482)
==================

========== ====== ALERT ======
Flow: ('192.168.50.140', 52550) -> ('34.107.221.82', 80)
Prediction: BENIGN (conf=0.5482)
==================

========== ====== ALERT ======
Flow: ('192.168.50.140', 8080) -> ('192.168.50.33', 41076)
Prediction: DDoS (conf=0.7714)
==================
```

Paper Submission Status

Conference-Name

International Conference on Smart Systems, Cyber Security & Communication (ICSSCC 2026)

Publication

Springer – Lecture Notes in Networks and Systems

Conference Type

- Peer-reviewed International Conference
- Scopus Indexed (Springer Proceedings)

Conference Website

https://icsscc.in.net/

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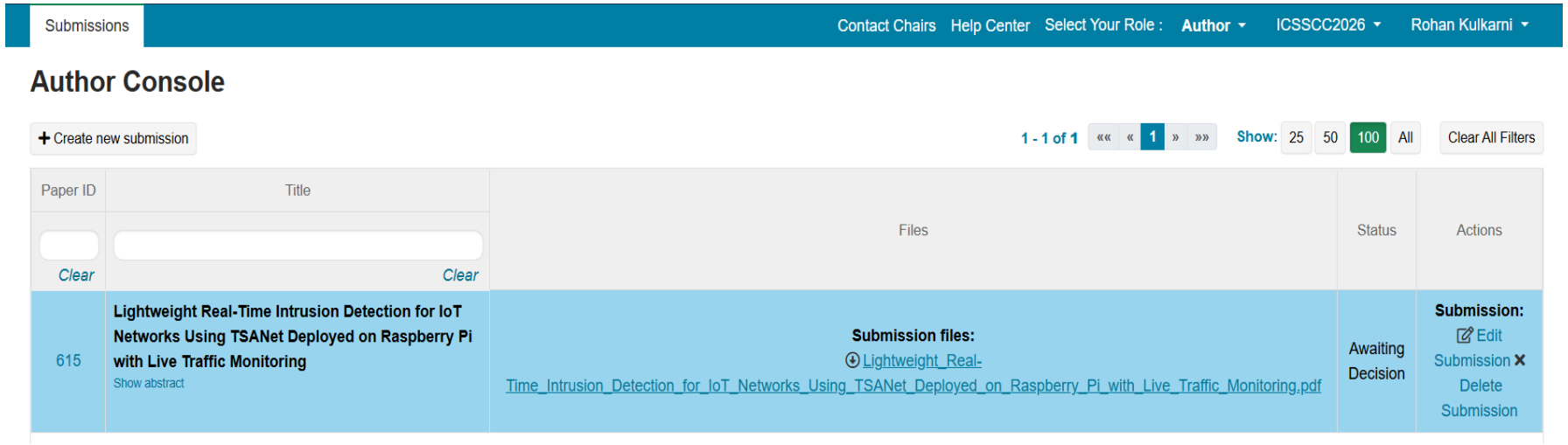
- Paper ID: 615
- Title: Lightweight Real-Time Intrusion Detection for IoT Networks Using TSANet Deployed on Raspberry Pi with Live Traffic Monitoring

Current Status

- Submitted successfully



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615	Lightweight Real-Time Intrusion Detection for IoT Networks Using TSANet Deployed on Raspberry Pi with Live Traffic Monitoring Show abstract	Submission files: Lightweight_Real-Time_Intrusion_Detection_for_IoT_Networks_Using_TSANet_Deployed_on_Raspberry_Pi_with_Live_Traffic_Monitoring.pdf	Awaiting Decision	Submission: Edit Submission Delete Submission

Paper Submission Ack

Submission Summary

Conference Name
International Conference on Sustainable Smart Computing and Communications

Paper ID
615

Paper Title
Lightweight Real-Time Intrusion Detection for IoT Networks Using TSANet Deployed on Raspberry Pi with Live Traffic Monitoring

Abstract
Intrusion Detection Systems (IDS) are essential for protecting IoT networks; however, many deep learning–based IDS solutions are too computationally intensive to be deployed on resource-constrained edge devices. This paper proposes TSANet , a lightweight Temporal–Spatial Attention Network for real-time intrusion detection on low-power hardware such as the Raspberry Pi 5. The proposed model integrates a 1D Convolutional Neural Network for spatial feature learning, an LSTM network for temporal sequence modeling, and attention mechanisms to highlight important traffic features and time steps. TSANet is trained and evaluated on the CICIDS2017 dataset and optimized using pruning and quantization-aware training to reduce model size and inference latency. Experimental results show that the optimized model achieves 98.3% offline accuracy and 96.8% real-time detection accuracy on Raspberry Pi, with an inference latency of 8–12 ms per flow, demonstrating its effectiveness for practical IoT edge based intrusion detection.

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Authors
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Ryan Dias (KLE TECHNOLOGICAL UNIVERSITY) <ryandias.mss@kletech.ac.in>

Submission Files
Lightweight_Real-Time_Intrusion_Detection_for_IoT_Networks_Using_TSANet_Deployed_on_Raspberry_Pi_with_Live_Traffic_Monitoring.pdf (822.8 Kb, 12/27/2025, 11:50:46 AM)

Submission Questions Response
1. Springer Template
Should the paper be formatted strictly according to the Springer template? Strictly follow the Springer manuscript preparation guidelines https://www.springer.com/gp/authors-editors/book-authors-editors/your-publication-journey/manuscript-preparation

Agreement accepted

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Paper 615 summary External Inbox ×

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to me

11:51 AM (2 minutes ago)

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Track Name: ICSSCC2026

Paper ID: 615

Paper Title: Lightweight Real-Time Intrusion Detection for IoT Networks Using TSANet Deployed on Raspberry Pi with Live Traffic Monitoring

Abstract:
Intrusion Detection Systems (IDS) are essential for protecting IoT networks; however, many deep learning-based IDS solutions are too computationally intensive to be deployed on resource-constrained edge devices. This paper proposes TSANet , a lightweight Temporal-Spatial Attention Network for real-time intrusion detection on low-power hardware such as the Raspberry Pi 5. The proposed model integrates a 1D Convolutional Neural Network for spatial feature learning, an LSTM network for temporal sequence modeling, and attention mechanisms to highlight important traffic features and time steps. TSANet is trained and evaluated on the CICIDS2017 dataset and optimized using pruning and quantization-aware training to reduce model size and inference latency. Experimental results show that the optimized model achieves 98.3% offline accuracy and 96.8% real-time detection accuracy on Raspberry Pi, with an inference latency of 8-12 ms per flow, demonstrating its effectiveness for practical IoT edge based intrusion detection.

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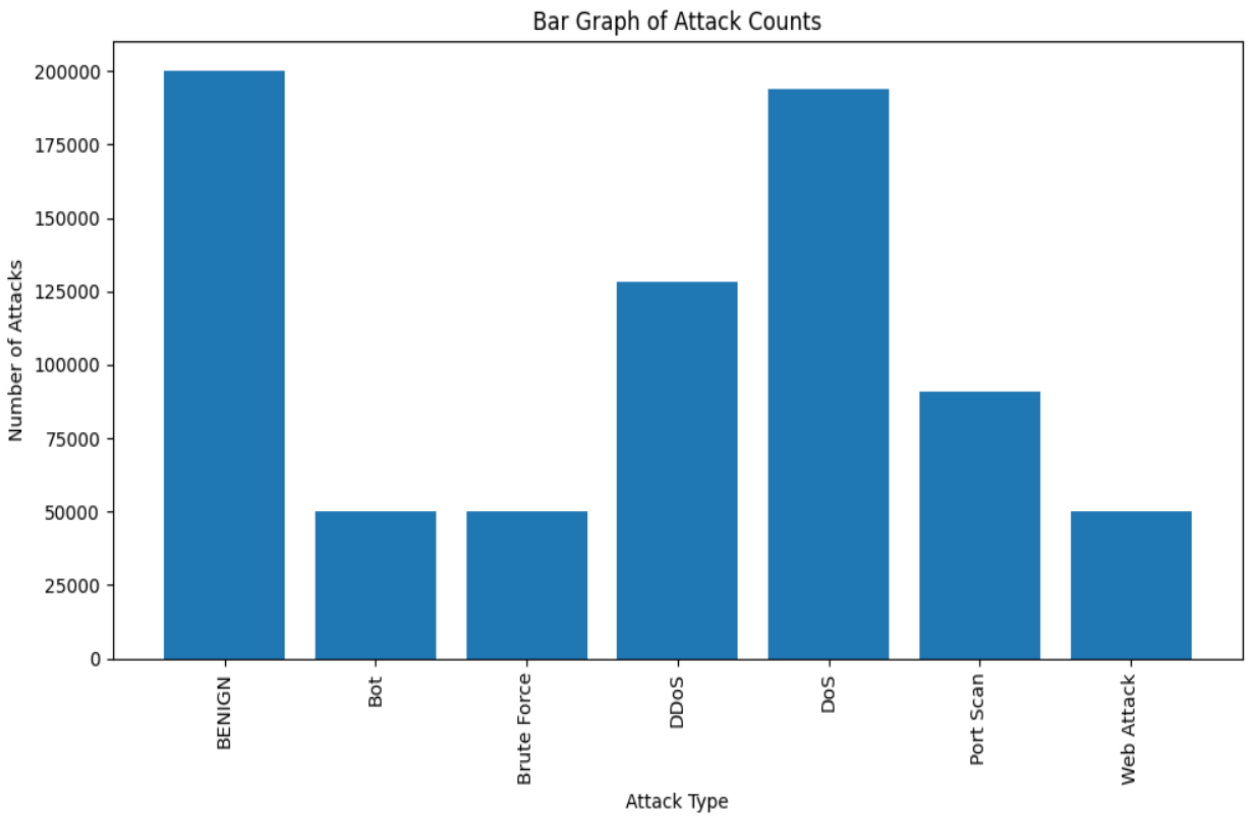
Q&A

THANKYOU

IMPLEMENTATION.

Dataset Preparation

- Collected and preprocessed the CICIDS-2017 dataset for intrusion detection tasks.
- Provides realistic, labeled network traffic for developing and testing Intrusion Detection Systems.
- Performed feature selection and normalization to prepare data for model training.
- Real users performed activities like web browsing , emails , chats, file transfers, video streaming, VoIP.
- Attack scenarios were simulated (DoS, DDoS, BruteForce , Botnet, SQL Injection, XSS, Infiltration, Heartbleed).
- Every record is tagged as either Benign or a Specific Attack Type.
- Makes it suitable for training deep learning models.



IMPLEMENTATION.

Model Design: TSANet

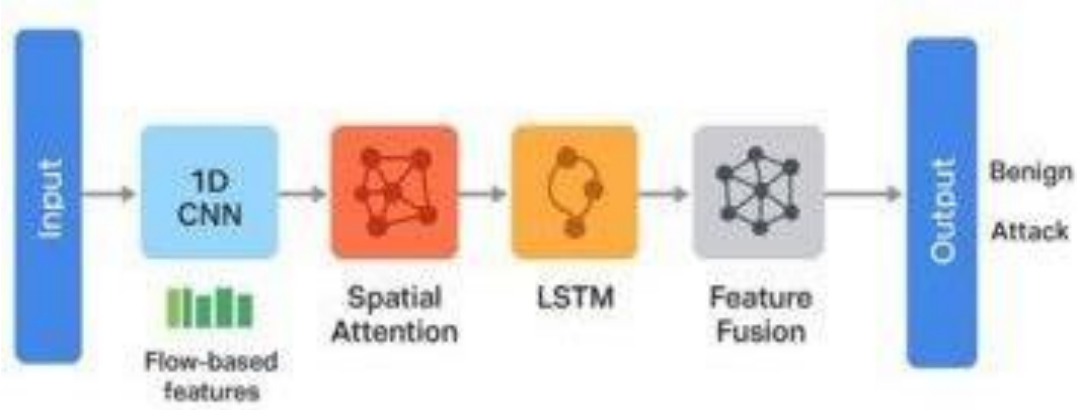
- A **hybrid deep learning model** designed for intrusion detection.
- Combines **CNN + Attention + LSTM** for capturing both spatial and temporal patterns in network traffic.

- **CNN** → extracts local/spatial patterns between features.

- **Attention** → focuses on the most relevant features and times.

- **LSTM** → models how traffic changes **over time**.

This combo is powerful for **intrusion detection** because attacks have both:
What features matter (spatial)
When they happen (temporal).



IMPLEMENTATION.

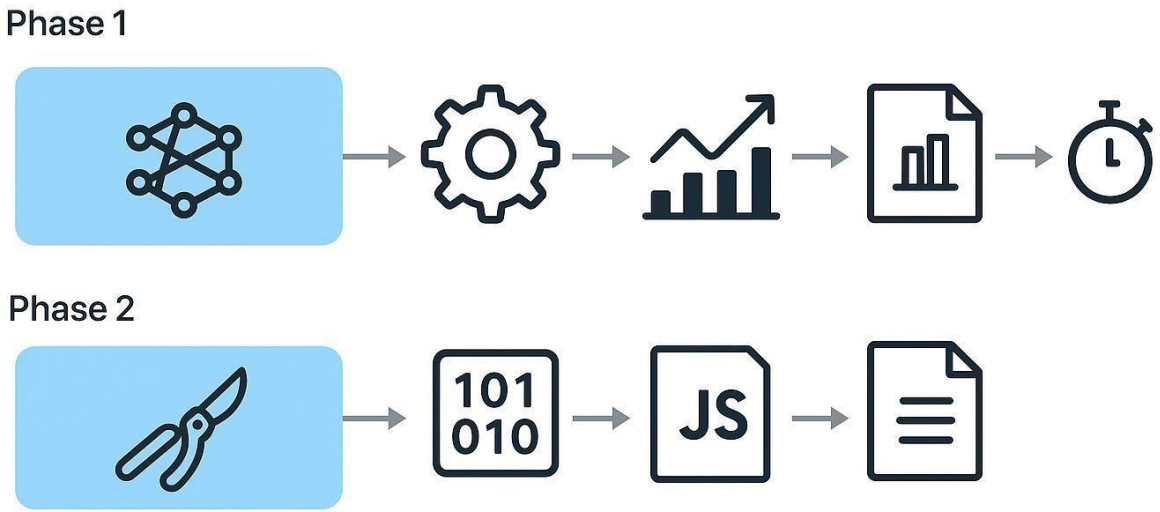
ModelTraining

Phase1:FullFeatureTraining

- **Dataset:**CIC-IDS2017(32features).
- **Learning Rate**=0.001.
- **BatchSize:**256,Epochs:30.
- **Loss Function:** Cross-Entropy Loss (multi-class classification).

Phase2:ModelOptimization

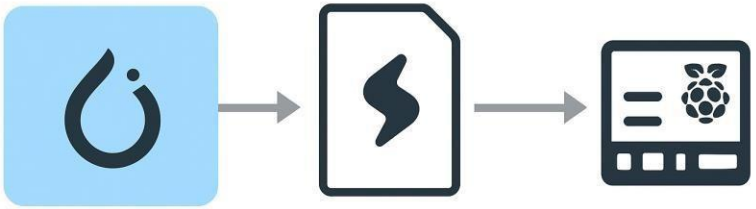
- **Pruning:**Removed redundant neurons→reduced Model size, improved inference speed.
- **Quantization-aware Training:** Prepared for low-precisionoperations(int8/float16).
- **TorchScript Conversion:** For lightweight deployment(onedgedeviceslikeRaspberryPi).



IMPLEMENTATION

Model Conversion for Deployment

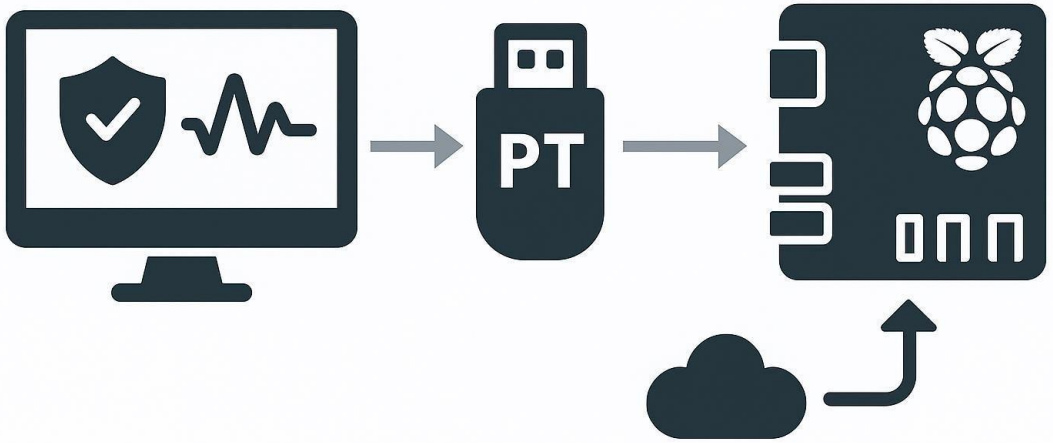
- Converted the trained PyTorch model into **TorchScript** format.
- Optimized the model for deployment on **RaspberryPi** with limited resources.
- Raw PyTorch models are **too heavy for Raspberry Pi**
- Needed a **lightweight, portable format** to ensure **low-latency real-time detection**.
- Solution: **Optimize + Convert to TorchScript** for deployment.



IMPLEMENTATION

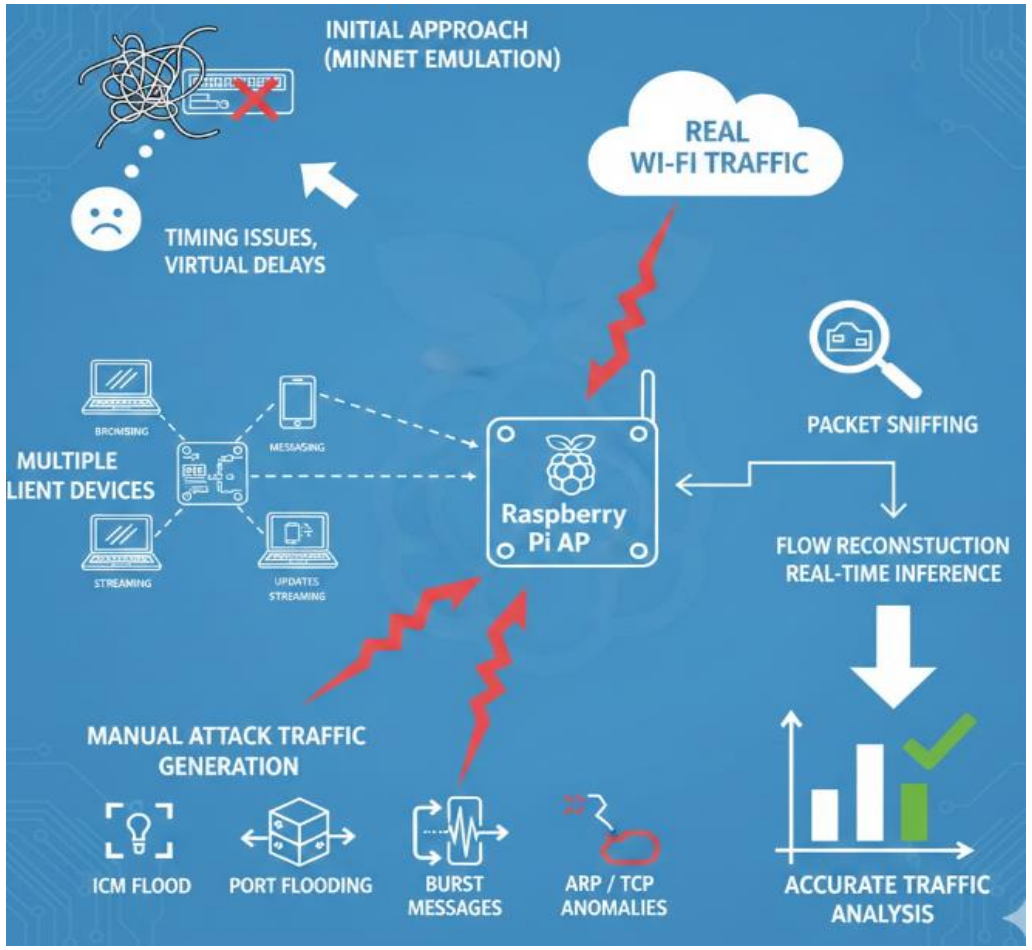
Deployment on Raspberry Pi

- Deployed the TorchScript IDS model on to the Raspberry Pi.
- Integrated real-time packet capturing for live intrusion detection.
- Model Transfer Trained and optimized TorchScript model (.pt file) was **transferred via USB (pendrive)** to Raspberry Pi.
- File placed in the working directory (/home/pi/IDS/
- Raspberry Pi Environment Setup Installed required dependencies: **Python 3.x** , PyTorch , **Scapy / Socket libraries** for traffic feature extraction. TorchVision C NumPy



IMPLEMENTATION

- **Initial Approach (Mininet Emulation)**
  Attempted to generate synthetic traffic using Mininet hosts and virtual switches. However, due to timing issues, virtual packet delays, and incomplete flow formation, Mininet could not provide reliable real-time packets for Raspberry Pi sniffing.
- **Final Approach (Real Wi-Fi Traffic Through Raspberry Pi AP)**
  The Raspberry Pi was configured as a **wireless access point** using *hostapd* and *dnsmasq*.
  Multiple laptops and mobile devices connected to this AP and generated **natural, real-world traffic** such as browsing, messaging, updates, and streaming.
- **Manual Attack Traffic Generation**
  To evaluate model behavior, controlled attacks were triggered from client devices:
  – ICMP flood (ping flood)
  – Port-based flooding
  – Burst-style client–server messages
  – ARP / TCP anomalies during stress testing
- **All traffic passed through the Pi**, enabling accurate packet sniffing, flow reconstruction, and real-time inference.



Empathy

- IoT devices are becoming a central part of everyday life—homes, campuses, industries, and personal devices continuously stay connected to Wi-Fi networks. However, most users are unaware that these devices are highly vulnerable to cyberattacks such as DoS, DDoS, brute force attempts, and other network intrusions. In typical home or small-network environments:
  - Users depend on **default Wi-Fi routers** that provide almost no intelligent threat monitoring.
  - Attacks often go **unnoticed** until the network slows down, devices malfunction, or data is stolen.
  - Cloud-based IDS solutions introduce **latency, privacy concerns, and dependency on internet connectivity**.
  - Meanwhile, IoT traffic is unpredictable and noisy, making manual analysis impossible for regular users.
  - This project aims to **stand in the user's shoes** and solve the real pain-point

BACKGROUND MOTIVATION

Modern IoT networks—smart homes, campus networks, industryautomation—generate continuous traffic from sensors, mobile devices,and embedded systems. As IoT adoption increases, the number ofcyberattacks targeting these networks has also grown rapidly..

This project is driven by the need to build a **practical, real-time, low-cost IDS** that can run directly inside an IoT network.

LITERATURE REVIEW

3. Paper Title : Deep Learning in IoT Intrusion Detection

- **Survey-based methodology**: Systematic review of ~100 papers. **Classification by model types**: DNNs, CNNs, RNNs (LSTM/GRU), CNN-RNN hybrids, Boltzmann Machines, DBNs, Autoencoders, GANs. Literature search using Scopus with IoT, deep learning, and IDS criteria.
- **Multiple benchmark datasets**: NSL-KDD (148,517 samples), ISCX 2012 (500k+ samples), UNSW-NB15 (257,673 samples), CIDDS-001 (2M+ samples), CICIDS 2017 (618,976 samples). **Challenge**: Outdated datasets not reflecting modern IoT attacks.

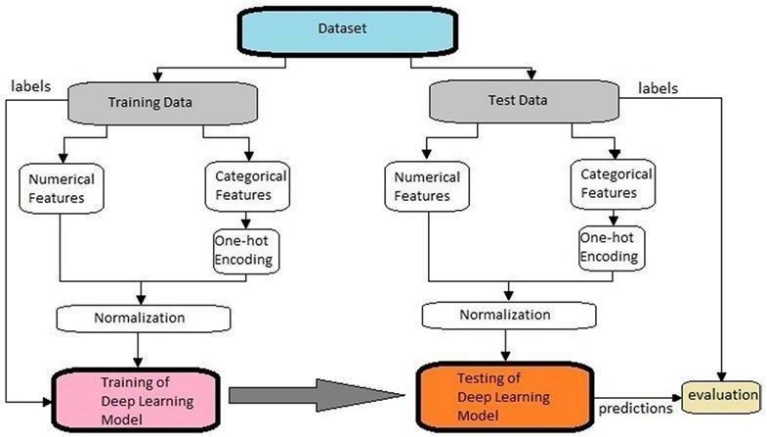


Fig. 5 Model training and testing

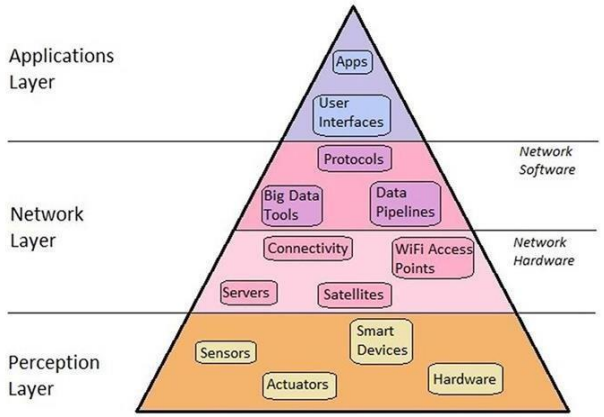
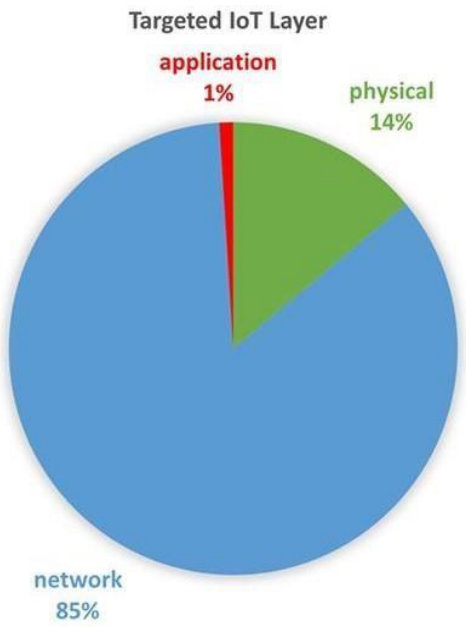
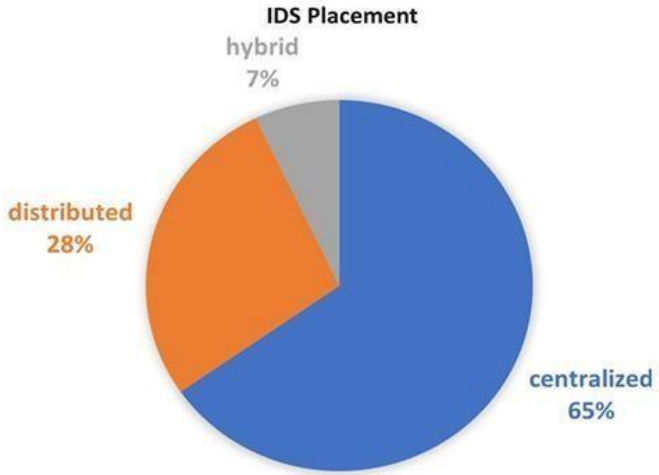
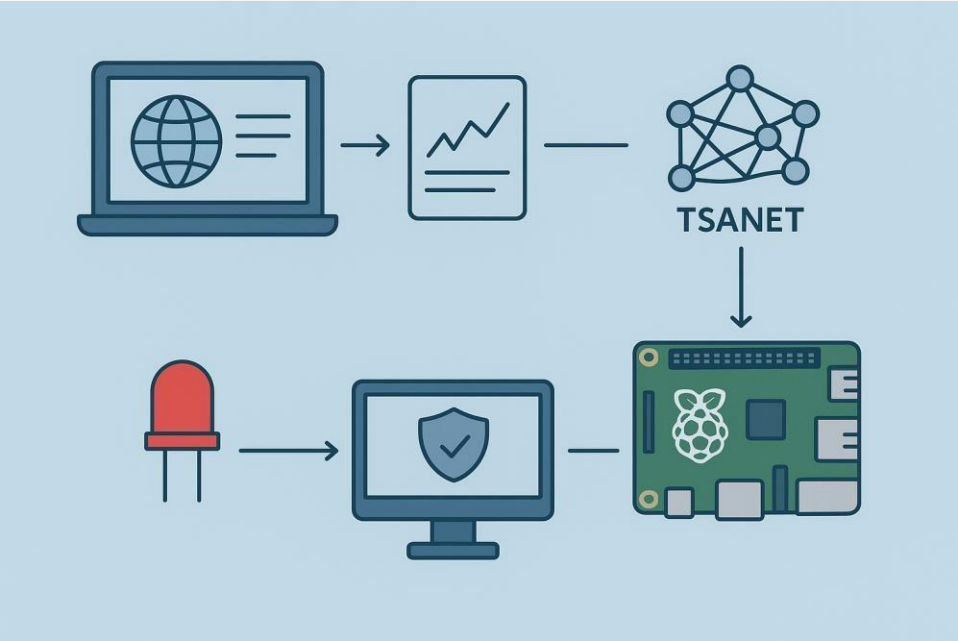


Fig. 2 IoT architecture

- **Accuracy ranges**: CNNs (99.13% ISCX2012), RNNs (99.99% ISCX2012), CNN-RNN hybrids (99.92%), DBNs (99.45% NSL-KDD), Autoencoders (99.96% NSL-KDD). **Additional metrics**: FPR, DR, Precision, Recall, F-measure reported.
- **Real-world validation**: Gap between simulated and actual IoT environments. **Standardization**: Lack of consistent benchmarks and evaluation metrics.
- **Scalability**: DL deployment is challenging on resource-constrained IoT devices.







REQIUREMENTANALYSIS

In requirement analysis, we describe what the system should do and the conditions it should meet.

Functional Requirements

- Capture and process network traffic flows in real time.
- Extract 32 flow-level features from traffic.
- Classify traffic as *benign* or *attack* using TSANet model.
- Trigger LED alert on Raspberry Pi when attack is detected.

Non-Functional Requirements

- Real-Time Performance: Low-latency detection suitable for edge environments.
- Lightweight Deployment: Model optimized to run on Raspberry Pi.
- Scalability: System adaptable to different IoT attack types and traffic volumes.
- Usability: Easy integration with IoT/edge devices without complex setup.

Paper Submission Status

Journal Name

International Journal of Networking and Virtual Organisations (IJNVO)

Publisher

Inderscience Publishers, United Kingdom

Journal Type & Indexing

- Peer-reviewed international journal
- Indexed in Scopus (Q4 category)

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- Manuscript ID: **IJNVO-303926**
- Title: *Lightweight Real-Time Intrusion Detection for IoT Networks Using TSANet Deployed on Raspberry Pi with Live Traffic Monitoring*

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- Submitted successfully
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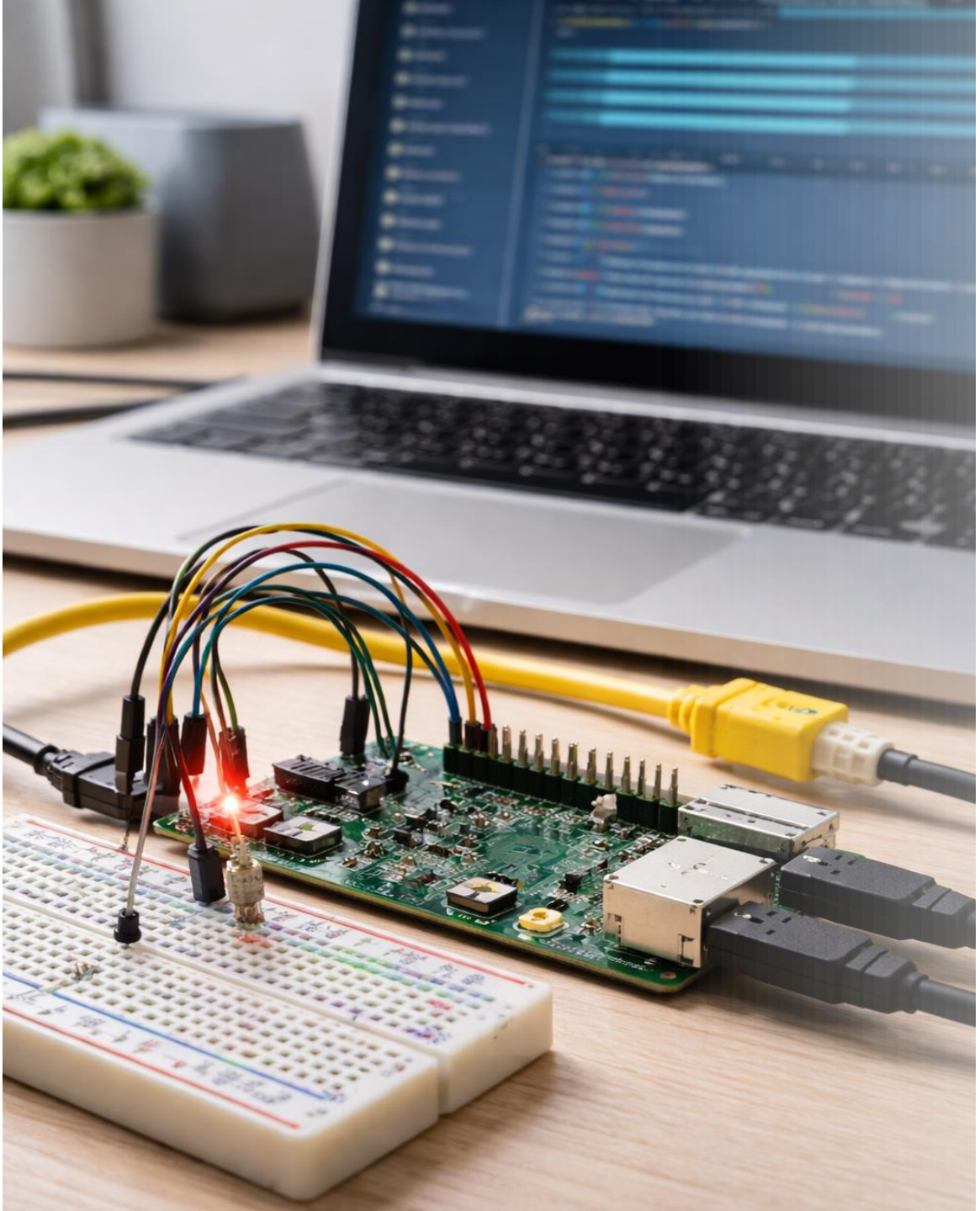

HARDWARE